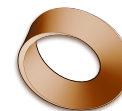


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.5 – Shadows & Eclipses

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can define **umbra**, **penumbra** and **antumbra** in relation to light sources and opaque objects.
- b) I can explain the alignment of the Sun, Earth and Moon required for solar and lunar eclipses.
- c) I can describe why eclipses do not occur every month and why the Moon appears red during a total lunar eclipse.

Theory

When an opaque object blocks light from a source, it creates a shadow with distinct regions. The **umbra** is the central, darkest part of the shadow where the light source is completely hidden. If you stand in the umbra, you see no part of the light source. Surrounding the umbra is the **penumbra**, a lighter, partial shadow where the light source is only partly blocked. Beyond the tip of the umbra lies the **antumbra**, a region where the opaque object appears smaller than the light source, allowing a ring of light to be visible around it.

A solar eclipse occurs when the Moon passes directly between the Sun and the Earth, casting its shadow on our planet. Observers in the umbra experience a total solar eclipse, while those in the penumbra see a partial eclipse. If the Moon is too far from Earth to completely cover the Sun, an **annular** eclipse occurs, creating a 'ring of fire' effect because the observer is standing in the antumbra.

A lunar eclipse happens when the Earth passes between the Sun and the Moon, casting its shadow onto the Moon. This can only happen during a full moon. However, eclipses do not occur every month because the Moon's orbit around the Earth is tilted by about 5 degrees relative to the Earth's orbit around the Sun. Most of the time, the Moon passes above or below the Earth's shadow rather than through it.

During a total lunar eclipse, when the Moon is fully within the Earth's umbra, it does not disappear completely but often turns a deep red colour. This happens because the Earth's atmosphere scatters shorter wavelengths of light (like blue) and refracts longer wavelengths (like red) towards the Moon. The Moon acts like a screen reflecting this filtered sunlight back to us on Earth.

Word bank: umbra · penumbra · antumbra · tilted · atmosphere · annular

Activity

Modelling Shadows with a Torch

You will need a bright torch (representing the Sun), a ball (representing the Moon or Earth), and a white wall or screen. Work in a darkened room.

- Hold the torch steady at one end of the table. Place the ball halfway along the table.
- Shine the torch onto the ball. Observe the shadow cast on the wall/screen behind the ball.
- Identify the darkest central part of the shadow (the umbra) and the lighter outer edge (the penumbra).
- Move the ball closer to the torch, then further away. Record how the size of the umbra changes in the table below.

Position of Ball	Size of Umbra (Small/Large)	Is the Shadow Sharp or Fuzzy?
Close to torch		
Middle distance		
Close to wall		

- Now, imagine the ball is the Earth and the screen is the Moon. Move the 'Moon' (screen) so it just touches the edge of the dark shadow. What type of eclipse would this represent?

Questions

1. Name the region of a shadow where the light source is completely blocked.

2. Explain why a solar eclipse can only happen during a new moon, whereas a lunar eclipse happens during a full moon.

3. Why don't we have a solar and lunar eclipse every single month?

4. During a total lunar eclipse, sunlight passes through Earth's atmosphere before reaching the Moon. Use your knowledge of light scattering to explain why the Moon looks red rather than blue or green.

Extension

Research the term 'saros cycle'. How long does it take for eclipses to repeat in a similar pattern, and why is this period so complex?
